

lab3

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The Data

The data we have are counts of deaths per episode of Game of Thrones.

```
GoT <- read_xlsx("GoT_Deaths.xlsx", col_names = T)
head(GoT)
```

1. Plot a histogram of the death counts.

```
hist(GoT$Count,
     main = "Histogram of Death Counts Per Episode",
     col = "skyblue",
     border = "white",
     xlab = "Death Counts")
```

The sampling distribution looks like this, centered around 3 to 4 deaths, roughly bell-shaped.

```
y <- GoT$Count
n <- length(y)
# Finish: obtain empirical mean
ybar <- mean(y)
sim_dat <- rpois(n, ybar)
qplot(sim_dat, bins = 20, xlab = "Simulated number of deaths",
      fill = I("#9ecae1"))
```

Plotting the histogram of the data next to the histogram of the sampled data using $\hat{\lambda} = \bar{y}$, we see that there are many more 0-valued observations in the observed data than there are in the simulated data. We might suspect that the Poisson model will not be a good fit for the data. For now, let's proceed using the Poisson model and see how it behaves.

```
# create data frame for side-by-side histograms
df <- rbind(data.frame(y, "observed") %>% rename(count = 1, data = 2),
            data.frame(sim_dat, "simulated") %>% rename(count = 1, data = 2))
ggplot(df, aes(x = count, fill = data))+
  geom_histogram(position = "dodge", bins = 20) +
  scale_fill_brewer() +
  labs(x = "Number of deaths", y = "Count")
```

Poisson Model

We have learned that the Gamma distribution is conjugate for Poisson data. Here, we have chosen the prior $\lambda \sim \text{Gamma}(10, 2)$ based on prior knowledge that Game of Thrones is a show filled with death. We run the model in Stan. We save the posterior sampled values of λ in the variable `lambda_draws`.

```
stan_dat <- list(y = y, N = n)
fit <- stan("lab-03-poisson-simple.stan", data = stan_dat,
           refresh = 0, chains = 2)
lambda_draws <- as.matrix(fit, pars = "lambda")
```

2. Use the function `mcmc_areas()` to plot the smoothed posterior distribution for λ with a 90% Highest Posterior Density region. Changing the `prob` parameter in `mcmc_areas()` will change the amount of probability mass that is highlighted. The highlighted region will begin in regions of highest density, and will move towards regions of lower density as `prob` gets larger.

```
mcmc_areas(lambda_draws, prob = 0.9)
print(fit, pars = "lambda")
```

The line in the plot represents the posterior mean, and the shaded area represents 90% of the posterior density. How does the posterior mean for λ compare to the sample mean?

The posterior mean for λ is very similar to the sample mean, which is both very close to 4. This makes sense, since the posterior mean is a weighted average of the prior mean and the sample mean, and we only had $\beta = 2$ for the prior but 73 episodes for the data. So the posterior mean is dominated by the sample mean.